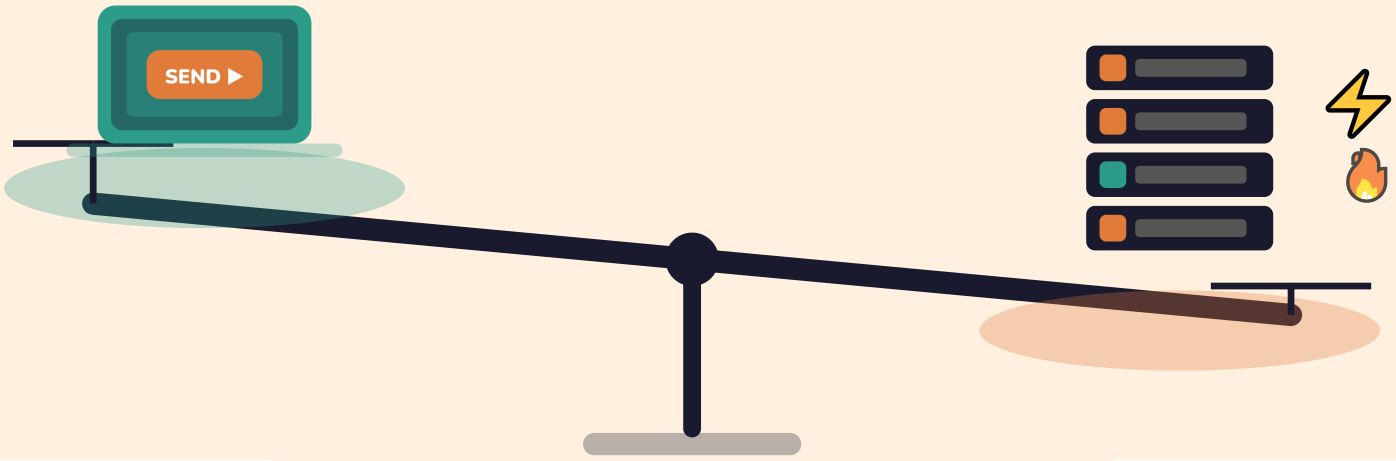
DOES AI USE ENERGY?

What is the environmental cost of a single AI query?



YOUR QUERY

→ data centre
→ electricity
→ water cooling
→ CO_2

SERVERS WEIGH MORE

SERVER SIDE

24/7 power
GPU clusters
cooling water
× billions of queries / day

RANK

CALCULATE

COMPARE

DOES AI USE ENERGY?

Students rarely connect digital actions to physical infrastructure. This card makes the chain concrete: a query → data centre → electricity → water cooling → CO_2. The goal is not guilt but informed awareness.

MAIN ACTIVITY

1. Introduce the energy chain: what physically happens when you send a message to an AI. Show that data centres use electricity 24/7 — even when no one is asking questions.
2. Give groups a list of 8 AI tasks (generating an image, writing an email, translating a sentence, running a climate model...). Students rank them by estimated energy cost, highest to lowest.
3. Reveal the real estimates (order of magnitude). Were any surprises? Compare each group's rankings.
4. Design a "digital energy label" for AI tasks — like a food nutrition label, but for energy use.

MATERIALS

- Printed AI task list (8 tasks per group)
- Ranking strips (cut and sort)
- Energy comparison reference cards
- Blank "energy label" template
- Reveal answer sheet (teacher holds)

REFLECTION

- Should we think about energy before using AI tools?
- Who is responsible — users, companies, or governments?
- What human activities use similar amounts of energy?
- Could AI ever be used to reduce energy use?

REALITY CHECK

Training a large AI model can produce as much CO_2 as five transatlantic flights.

Everyday use is far smaller, but adds up across billions of users. Data centres already consume about 1–2% of global electricity — and this is growing rapidly as AI use increases.

DIGITAL GAME

Carbon Counter

Scan to play →

⏱ 45 MIN | GROUP RANKING | INTERMEDIATE | CARD 13